

SCI 1399 Spectroscopy of Dyes

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Cyanine dyes are organic molecules with a polymethine chain in the center and nitrogen-containing rings on the sides. Polymethine chains are conjugated hydrocarbon chains, consisting of an alternating sequence of single and double bonds, with overlapping p orbitals forming π bonds. As seen in the "Particle in a Box" quantum model, longer polymethine chains reduce spacing between energy levels of the molecule, decreasing the gap between the Highest Occupied Molecular Orbital (HOMO) and the Lowest Unoccupied Molecular Orbital (LUMO). According to the Planck-Einstein relation, $E = \frac{hc}{\lambda}$. Since energy and wavelength are inversely related, a cyanine dye with a smaller HOMO-LUMO gap should absorb a higher wavelength of light. This behavior was observed by comparing the absorption spectra of 1,1-Diethyl-2,2'-cyanine iodide (Cy3) and 1,1'-Diethyl-2,2'-carbocyanine iodide (Cy5), which have different polymethine chain lengths.

Because cyanine dyes are nonpolar, in polar solvents, molecules are held together by weak interactions called supramolecular assemblies to reduce unfavorable interactions with the solvent. From molecule monomers, to clusters, to nanotubes, to bundles, the dyes assemble. They also show birefringence since they direct the flow of energy and light along the length of the tube. This was behavior was replicated by aggregating cyanine dye in saltwater and observing it under an optical microscope.

1 Experimental Design

A UV/VIs spectrometer was used to measure the absorbance spectra of 25mL solutions of Cy3 and Cy5 diluted in ethanol with approximately 0.5A given a cuvette with a 1cm path length. After computation using Beer's law ($c = \frac{A}{\epsilon l}$) and the dilution equation ($C_1V_1 = C_2V_2$), we decided to dilute 1.25mL and 0.528mL of 0.000185M solutions of Cy3 and Cy5 into 23.75mL and 24.47mL of ethanol. Micro-pipettes were used to transfer the dyes from their containers to 30mL beakers due to their accuracy, especially for smaller quantities of solutions. Ethanol was added with disposable pipettes until the solution reached the 25mL line for ease and speed since we were working with a larger volume of

chemical. Still, graduated cylinders should have been used instead, since they have more precise measurement, and our final concentration of solutions would not be affected by human error. The solutions should have also been stirred to ensure that they were homogeneous. Separate disposable pipettes were then used to move the final solution into cuvettes as accuracy of measurement was not necessary for this step. The UV/Vis spectrometer was calibrated with a cuvette of ethanol. Overall, glassware and equipment were selected based on accuracy, size, and convenience.

An optical microscope was used to observe the aggregation of a premixed dye and saltwater solution that was evaporated over a glass microscope slide. Two pieces of double-sided tape were placed on the side to create a channel for our solution.

In general, our lab etiquette had room for improvement. Our working station was definitely not sterile, as we often touched other surfaces, skin, and clothing with our gloves. Additionally, we could have been more efficient with our procedure if we went over our steps together as a group before starting.

2 Results

2.1 UV/Vis Spectra of Cyanine Dye Monomers

Our initial hypothesis is proved correct by the results below. With more overlapping p orbitals, electrons are delocalized over a larger region, resulting in closer molecular energy levels, and a smaller HOMO-LUMO gap. When calculating the gaps for Cy3 and Cy5 experimentally, we get:

$$E = \frac{hc}{\lambda}$$

$$E_{Cy3} = \frac{1240 \text{ eV} \cdot \text{nm}}{524 \text{ nm}} = 2.37 \text{ eV}$$

$$E_{Cy5} = \frac{1240 \text{ eV} \cdot \text{nm}}{711 \text{ nm}} = 1.74 \text{ eV}$$

The observed difference between the HOMO-LUMO gaps is

$$\Delta E_{\text{experimental}} = 2.37 \text{ eV} - 1.74 \text{ eV} = 0.63 \text{ eV}.$$

Theoretically, the Particle in a Box model predicts the absorption wavelength according to

$$\lambda(\text{nm}) = 63.7 \frac{(p+3)^2}{p+4},$$

where p is the number of carbon atoms in the polymethine chain.

For Cy3, $p = 3$:

$$\lambda_{Cy3} = 63.7 \frac{(3+3)^2}{3+4} = 327.6 \text{ nm}$$

$$E_{Cy3} = \frac{1240 \text{ eV} \cdot \text{nm}}{327.6 \text{ nm}} = 3.79 \text{ eV}.$$

For Cy5, $p = 5$:

$$\lambda_{Cy5} = 63.7 \frac{(5+3)^2}{5+4} = 453.0 \text{ nm}$$

$$E_{Cy5} = \frac{1240 \text{ eV} \cdot \text{nm}}{453.0 \text{ nm}} = 2.74 \text{ eV}.$$

The theoretical difference between the HOMO-LUMO gaps is

$$\Delta E_{\text{theoretical}} = 3.79 \text{ eV} - 2.74 \text{ eV} = 1.05 \text{ eV}.$$

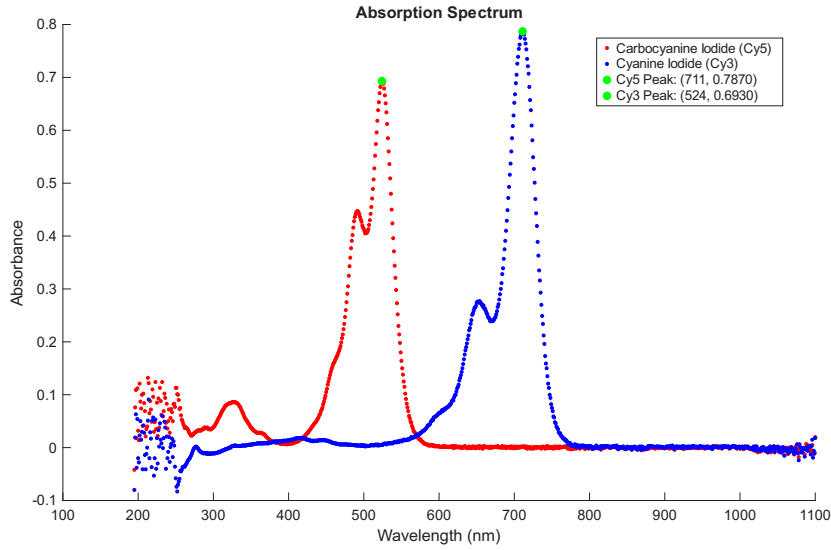


Figure 1: Absorption spectrum of Cy3 and Cy5 using UV/Vis.

Both dyes exhibited a smaller secondary absorption peak due to a vibrationally coupled electronic (or vibronic) transition. This is assumed because purely vibrational transitions occur at lower energies like infrared light, while the peaks seen in our results are in the visible/UV range, and thus, involve electronic excitation. This phenomenon is unlikely to be a result of incomplete mixing since

concentration affects absorbance intensity, not absorbance wavelength.

A band gap was seen both experimentally and theoretically. Therefore, the shift toward a longer wavelength indicates that less energy is required to excite an electron from the HOMO to LUMO. The PIB model, however, overestimates the energy gap because it assumes the electrons are completely confined to a fixed length L . Comparatively, real cyanine dye allows electron density to extend a little into the nitrogen-containing end groups, so the effective region where π electrons are delocalized is larger than assumed by the model and spacing between electronic energy levels is smaller.

2.2 Cyanine Dye Aggregate Nanotubes Imaged by Optical Microscope

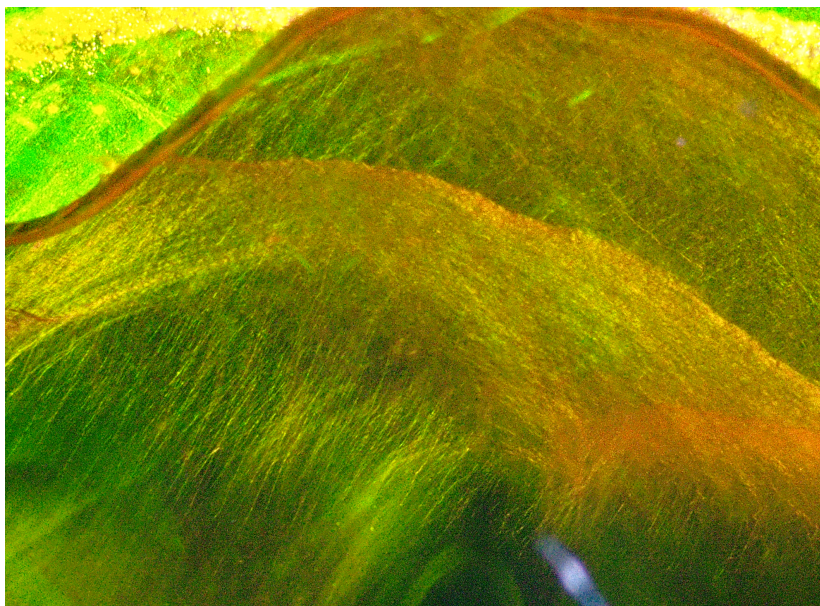


Figure 2: Low Resolution (229x Zoom) Nanotube Image

As shown by the two images we captured, we succeeded in creating a dense network of overlapping nanotubes. Unfortunately, because of an issue with the exported metadata, we were not able to obtain the scan size of the recorded images, leaving us unable to create a scale bar for them. Nonetheless, the variations in color throughout the images can be attributed to the fact that the orientation of each nanotube determines how light interacts with it, which is a phenomenon known as Birefringence. On a larger scale, this also suggests that the organization of the nanotubes can influence, and thus, be influenced

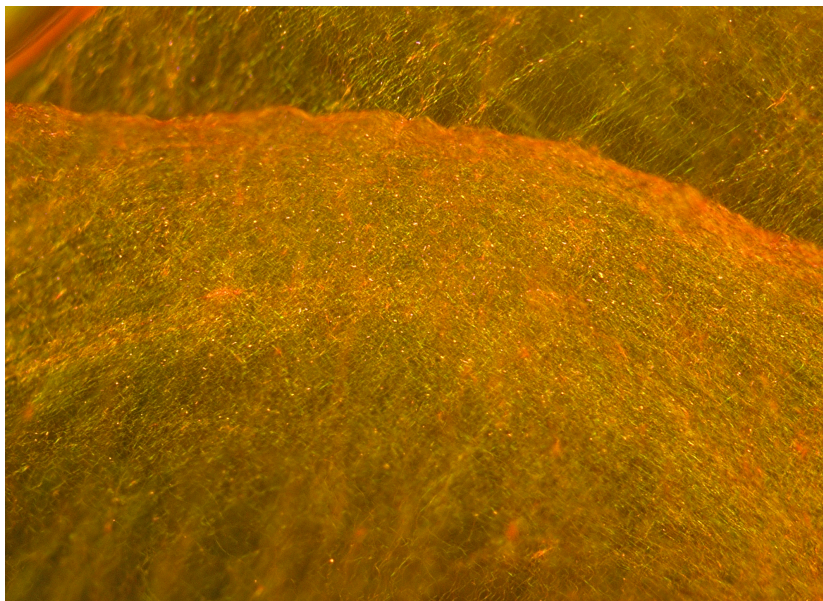


Figure 3: Higher-Resolution (502x Zoom) Nanotube Image

by, external environmental factors. For example, a change from salt water to another solution could potentially alter the orientation of the nanotubes and thus the observable properties of the material as a whole.

3 Next Steps

To further explore the aggregation of cyanine dyes, we could investigate which experimental parameters control the orientation of the dye's nanotubes. Evaporation rate, temperature, dye/salt concentration, and channel geometry could be varied, and the resulting nanotube alignment could be observed under an optical microscope. Since these nanotubes have direction-dependent optical properties, being able to manipulate their orientation could be useful for waveguiding light. Therefore, our next scientific question would be: can processing conditions alter nanoscale structure to change its optical properties?